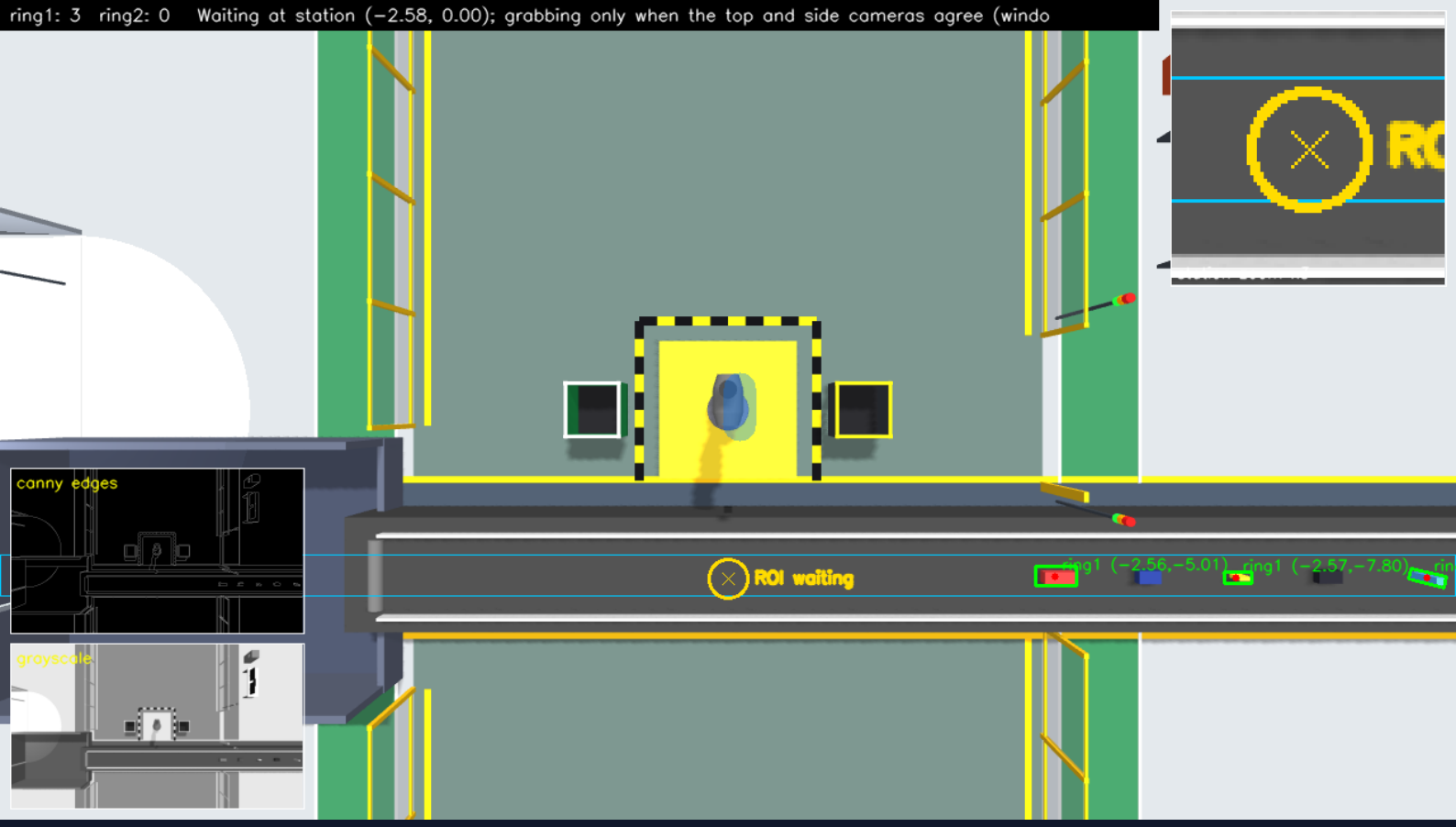


REPORT 03 · FACTORY BENT SIMULATION WITH A BCR ROBOT ARM

Vision Pipeline Evolution

From threshold scripts to three cross-checked cameras and pluggable OpenCV / YOLO detectors

ROS 2 Jazzy · Gazebo Harmonic · OpenCV · PyQt5 · BCR 7-axis arm

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1. Starting point: standalone scripts

The first vision code (March 2026) lived outside any ROS package. `camera_viewer.py` showed the bridged `/overhead_camera/image`, `object_detection_base.py` ran a threshold and contour pass, and `object_detection.py` added background subtraction, tracking, counting and size measurements in pixels, written to `measurement_logs/`. Nothing was published to ROS, and nothing converted pixels to world coordinates. The arm therefore could not use the results.

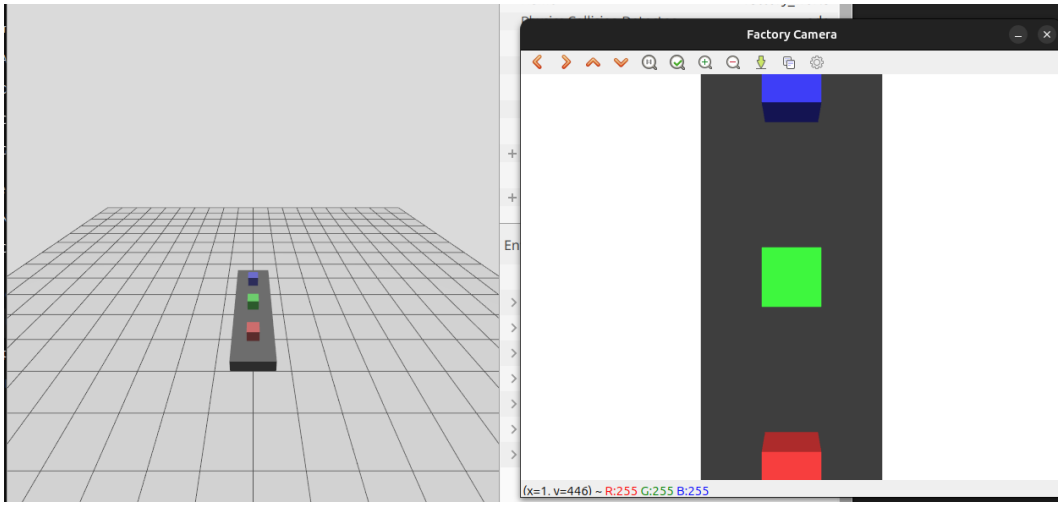


Figure 1. March 2026: the straight conveyor and the first camera window.

2. The ring detector (Oct 29)

`vision_geometry.py` models the overhead camera the way Gazebo does (the camera looks along +x, image right is −y, and image up is +z). With the camera 13 m up and pitched π/2, a pixel ray can be intersected with the plane of the product's top, which gives world x and y. `ring_object_detector` runs this pipeline on every frame:

```
BGR -> grayscale -> Gaussian blur
   -> intensity threshold  OR  filled Canny edges   (gray_margin, canny_low, canny_high)
   -> morphology (close/open)
   -> contours inside each ring's belt annulus      (other pixels ignored)
   -> minAreaRect -> centre, size, yaw
   -> pixel ray ∩ product-top plane -> world (x, y)
   -> /vision/ring1_objects, /vision/ring2_objects (PoseArray), /vision/detections (JSON)
```

The arm crosses the camera's view, so its pixels are masked out. `arm_mask_from_points` draws a thick polyline through the link frames it reads from TF. The tuning was checked against saved debug frames.

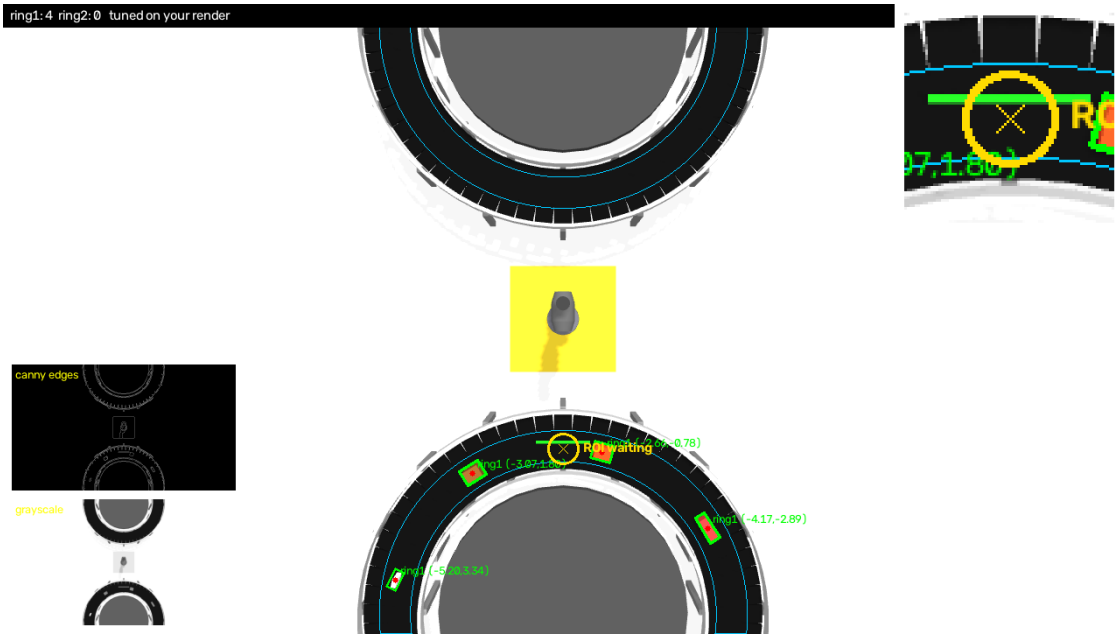


Figure 2. Tuned detection on both rings, with the zoom inset and Canny thumbnails.

3. Tracking moving products (station mode)

Products move at about 0.24 m/s, so a snapshot alone is not enough. `RoiTracker` follows every object near the catch station, estimates its speed along the ring, and predicts where it will be after `catch_latency` (0.35 s). The catch is triggered when that predicted position enters the ROI circle (`roi_radius` 0.30 m). The live window shows the ROI, the tracked object and a zoomed inset.

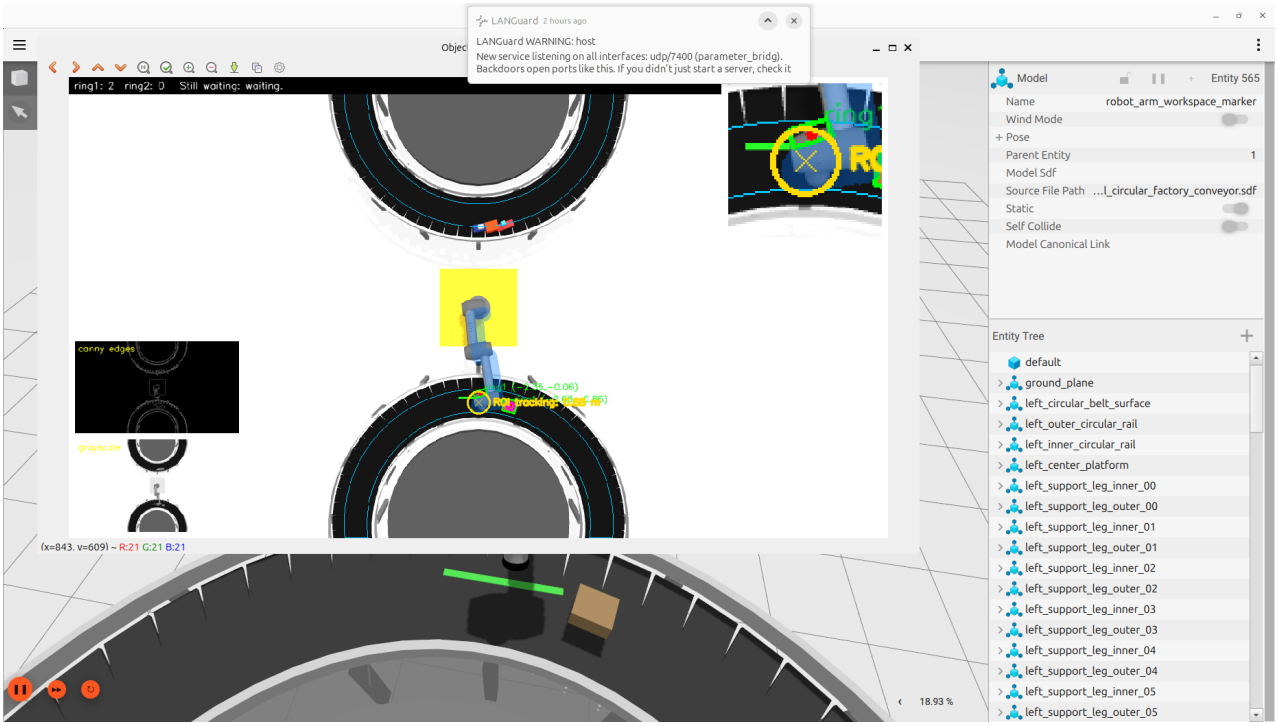


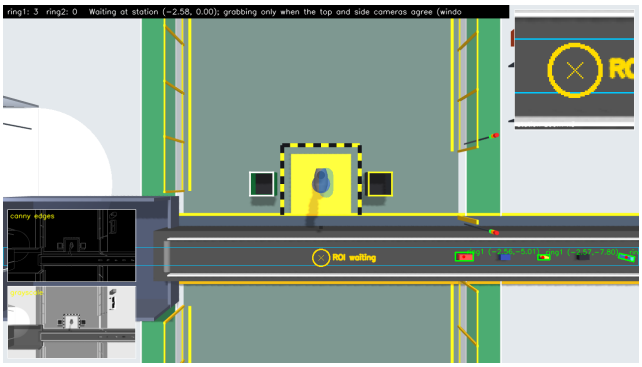
Figure 3. The live Object Detection window during a catch (Oct 30, 03:00).

4. A second opinion: the side camera

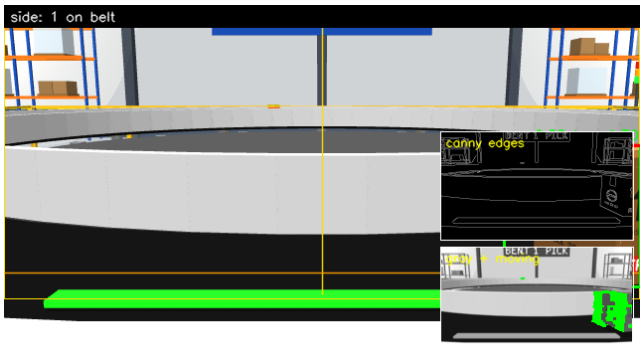
A single top view cannot tell a product standing on the belt from a shadow or the far side of the ring, and it cannot measure height. So a `side_camera` was placed low at (-1.50, 0, 1.80), looking straight at the station. `side_vision.py`:

```
grayscale -> blur -> MOG2 background subtraction (the side view is static except for products)
   -> Canny of the moving region -> morphology -> contours in the station band
   -> keep blobs whose bottom sits on the belt at the station plane
      (far-side products project higher and are dropped)
   -> y along the belt, top height z_top  ->  /vision/side_check
```

The controller grabs only when the overhead track and the side blob agree within `cross_check_tol` (0.20 m) and the side y is within `catch_window` (0.25 m) of the station. If they disagree, the product is let through. The measured height also tells the arm how far to descend and which movement style to use.



Overhead detections



Side camera cross-check

5. Is the drop zone free? The release camera

When bent 2 began to move (the ring2 phase), placing a product could hit one already riding there. `release_zone.py` learns the empty belt as the per-pixel median of the first frames, so passing objects drop out of it. Each frame is compared with it, thresholded, and edge-checked, with the arm masked out. The zone is EMPTY when no blob is large enough. The arm waits 0.8 m before the zone until it is empty.

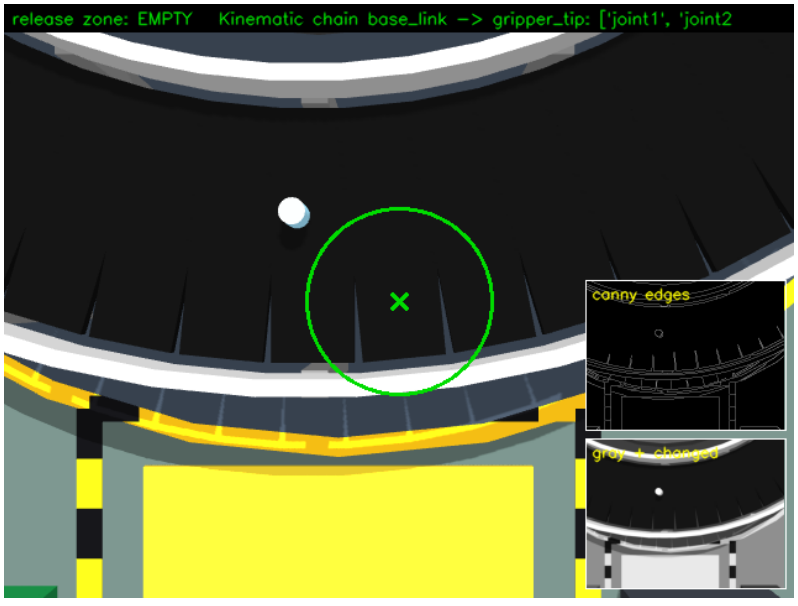


Figure 5. The Release Zone window: a green circle means empty.

6. Making it robust

- **Decor without false positives.** When the factory hall, fence and signs were added, the camera views were compared before and after, and nothing was placed over the belts.

- **Realistic products.** Labels and details were kept on the sides, and the top of each product keeps its catalogue colour, so the footprint and the colour classifier do not change. Nothing is saturated green, because green is the overhead camera's counting gate.
- **Custom products.** A user's colour is adjusted automatically if the detectors could not see it (near black, too light or strongly green).
- **Every world.** `WORLD_LAYOUTS` gives the detector the belt shape (annulus or straight lane) for each world.

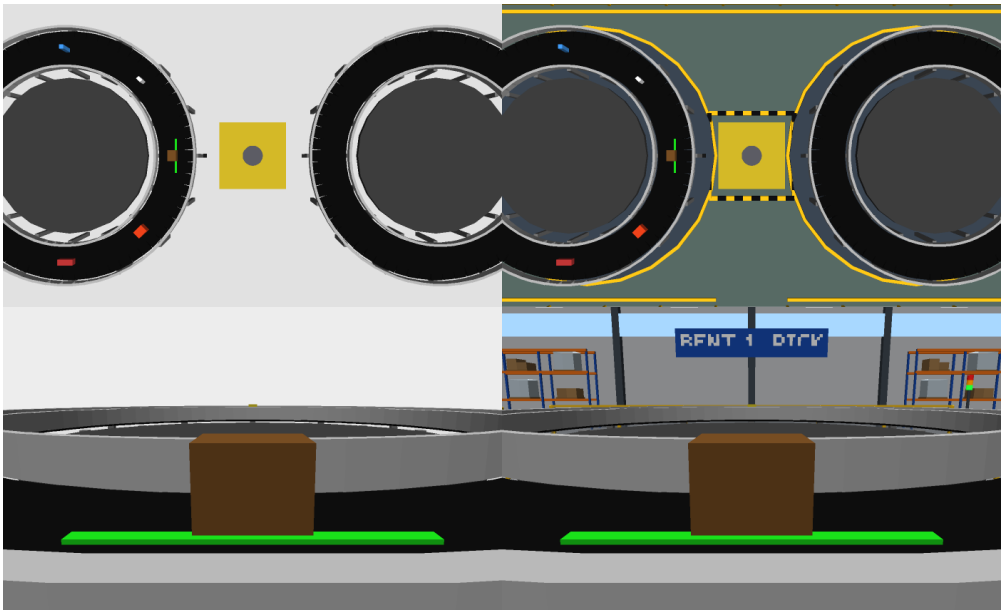


Figure 6. Camera views before and after the world decor.

7. From one detector to a detector slot: the vision lab

On Oct 31 the fixed detector became replaceable. Any class that returns pixel detections is converted to world coordinates, assigned to a bent, masked for the arm, and published by the lab. The built-in options are listed below.

Detector	Method	Labels
`opencv_contours`	the project's tuned grayscale + Canny contour detector	by colour (classify.py)
`opencv_color`	Lab colour segmentation against target product colours	target names
`opencv_bgsub`	MOG2 background subtraction	by colour
`yolo`	Ultralytics detect / OBB / segmentation weights	trained classes

COCO weights do not know the factory products, so lab 07 records an **auto-labelled YOLO dataset** while the arm works. The labels come from the OpenCV detector, which is checked against the simulation's ground truth. Products are only about 20 px wide in the 1280×720 image, so training uses `imgsz=1280`.

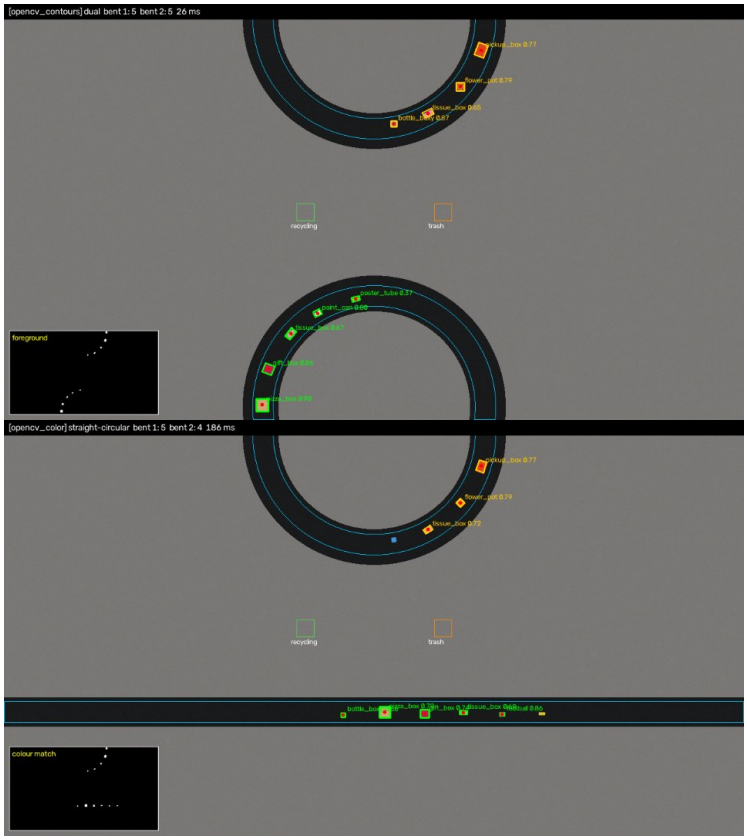


Figure 7. Offline scoring on synthetic scenes of every world (precision, recall, label accuracy, position error).

Summary

Stage	Cameras	Question answered
Mar 2026	overhead	Is something there? (pixels only)
Oct 29 vision	overhead	Where is it in the world?
Oct 29 station	overhead	When will it reach the arm?
Oct 29 sidecam	overhead + side	Is it really on the belt, and how tall is it?
Oct 30 ring2	+ release	Is the drop zone free?
Oct 31 lab	any + 12 station cameras	Which detector is best, and what is the product?